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Macroalgae (*Macrocystis integrifolia*)- derived biopolymer: A sustainable approach to bioplastic synthesis in the context of environmental impact assessment

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Abstract- The exponential rise of plastic production was a phenomenon of the 20th century on a historical scale. Plastics gained popularity due to their affordability and versatility, leading to widespread use across various industries. However, their non-biodegradable nature and harmful effects on the environment, animals, and human health prompted the development of bioplastics as an alternative. Bioplastics, derived from renewable biological sources, offer a solution to the issues posed by traditional plastics. Seaweeds have emerged as a promising candidate for bioplastic production due to their abundant biomass, ability to thrive in diverse environments, and natural cultivation processes compared to other microbial sources. Additionally, seaweeds offer cost-effectiveness, minimal disruption to the food chain, and independence from chemical inputs. This study focuses on synthesizing seaweed-based bioplastics with different seaweed compositions and plasticizers. Seaweed-based bioplastics exhibit enhanced resistance to microwave radiation, reduced brittleness, and increased durability compared to traditional plastics. The potential for replacing single-use plastics in food packaging and other applications is also explored. In summary, this review highlights the significance, benefits, and applications of seaweeds as a sustainable source for bioplastics.

Key words: Macroalgae, Biopolymer, Sustainable Approach, Macrocystis integrifolia, Carrageenan and Alginate

INTRODUCTION

The proliferation of petroleum-based plastics is driven by human demand and their unmatched versatility. However, these non-biodegradable materials persist in the environment for centuries, breaking down into harmful nano particles that infiltrate ecosystems and endanger marine life, wildlife, and potentially human health, including reaching human placenta. Seaweed-derived bioplastics offer an environmentally friendly, cost-effective, and non-toxic alternative. Leveraging seaweed's polymer composition, which is based on sugars, facilitates the creation of high-quality biodegradable plastics. Seaweed's abundance and widespread use in various scientific fields, such as food technology, microbiology, and biotechnology, further underscore its potential as a source for synthesizing bioplastics.[1]

The worldwide production capacity of bioplastics is estimated at 327,000 tonnes, while global consumption stands at approximately 12.3 million tonnes. This highlights a significant disparity between demand and production, indicating that the utilization of bioplastics is still in its early stages and does not adequately meet market needs.[2]

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Current Scenario of Plastic Waste in India

India ranks second globally in terms of population, trailing only China, with a population of 1.324 billion people. This substantial population size is driving industrialization and swift urban development in the country.[3] The Central Pollution Control Board (CPCB), in collaboration with NEERI, conducted a survey on solid waste management in 59 cities between 2004 and 2005, including 35 metro cities and 24 state capitals. Over the past few decades, there has been an annual increase in per capita waste generation ranging from 1% to 1.33%. If this trend persists, India is likely to witness a surge in waste generation from less than 40,000 tons per year to exceeding 125,000 tons by the year 2030.[4]

Seaweeds

Seaweeds, also known as macroalgae, typically inhabit coastal areas where they attach to rocks or other substrates. They are readily accessible, capable of thriving in diverse environments, cost-effective, and require no fertilizers or pesticides for cultivation. Seaweeds boast high biomass and are abundant in polysaccharides such as agarose, ulvan, and fucoidan.[1]

Seaweeds are categorized into three main groups according to their pigments and coloration.[2]

- Green seaweeds (Chlorophyta) - Cordium fragile
- Brown seaweeds (Pheophyta) - Macrocystis integrifolia
- Red seaweeds (Rhodophyta) - Porphyra

India boasts a coastline spanning over 7000 kilometers, which harbors a diverse range of seaweeds. Gujarat, in particular, exhibits significant diversity in both the Kutch and Khambhat Gulfs, owing to various coastal habitats influenced by factors such as coastal processes, geomorphology, and physiography.[5]

Seaweed-derived hydrocolloids, including alginate, agar, and carrageenan, serve as versatile biopolymers in bioplastic production. Polysaccharides, which are polymers, have garnered significant attention in the packaging industry. These polysaccharides, extracted from various species of red seaweed, encompass agar, alginate, and carrageenan.[1]

Extraction of Biopolymers

a. Carrageenan

Carrageenan is extracted from seaweed through a process involving soaking and alkaline treatment. The seaweed is soaked and then subjected to alkaline treatment at 60-65°C for 2 hours in an 8% KOH solution, with a seaweed to KOH solution ratio of 1:6 in the extraction tank. After treatment, the seaweed is rinsed with tap water until its pH reaches neutral. Carrageenan, primarily sourced from red seaweed, is a linear polysaccharide consisting of alternating galactose units linked by α-1,3-glycosidic and β-1,4-galactose bonds. It exhibits gelling, thickening, and stabilizing properties. Combining carrageenan with hydrophobic compounds enhances the compound's matrix.

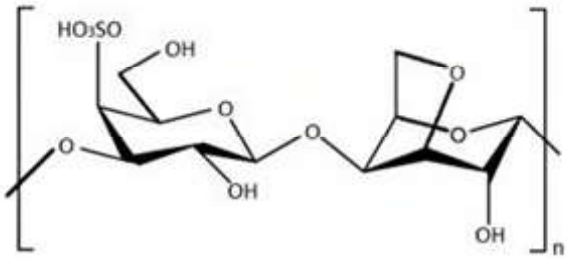


Fig.1- Chemical Structure of k-carrageenan Polymer

b. Alginate

Alginates are negatively charged linear poly saccharides abundant in brown seaweeds. Two primary methods are commonly used to recover alginate: the first involves inducing the formation of alginic acid by adding acid, while the second method entails adding calcium salt to obtain sodium alginate.

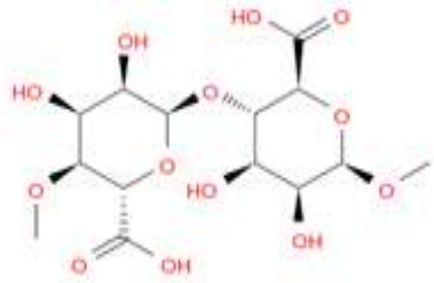


Fig. 2- Chemical Structure of Alginate Polymer

Alginate exhibits remarkable stabilizing and thickening properties and is highly hydrophilic, making it crucial to enhance its resistance to water by reinforcing the matrix.

The incorporation of calcium into the alginate matrix enhances stability and membrane resistance, presenting an intriguing avenue for developing biodegradable materials with antimicrobial properties and non-toxic packaging.

c. Agarose

Agarose contributes to the gelling ability of agar, rendering it highly valuable in pharmaceutical applications, where it also demonstrates outstanding film characteristics.

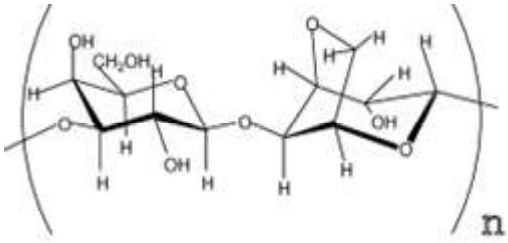


Fig. 3- Chemical Structure of Agarose Polymer

Agar exhibits low hygroscopicity, making it advantageous in packaging production. Additionally, agar films are biologically inert and readily interact with various bioactive substances and/or plasticizers, facilitating the formation of elastic and soft gels.

d. Advantages of Bioplastics

Bioplastics offer numerous benefits compared to traditional plastics, including biodegradability, a reduced carbon footprint, energy efficiency, versatility, distinctive mechanical and thermal properties, and societal acceptance. With significant potential, bioplastics can effectively substitute petroleum-based plastics across various sectors, from automotive to biomedical applications.[6] The utilization of bioplastics in food packaging offers significant advantages, including the reduction of carbon dioxide emissions and enhanced biodegradability.[7] Under controlled environmental conditions, microorganisms can break down biodegradable bioplastics without generating any toxic by-products.[6]

SAMPLE COLLECTION

Sampling Location (Mangrol Coast)

Latitude: 21°09'22.9"N, Longitude: 70°02'54.6"E, Mangrol is a town on the west coast of India in the Junagadh district in the state of Gujarat. Mangrol has a flat rocky intertidal belt, provided with many tide pools, gullies and crevices. The substratum there is made of limestone rock. The sandy beach gradually slops into the rocky inter tidal belt from landward side. Rhodophyceae shows more preponderance in the seaweed flora at Mangrol coast. It is found that in general, Chlorophyceae and Pheophycea observe during the September to January month while majority of Rhodophyceae found from January to March month.

Collection of Sample

On February 2nd, 2024, an expedition was undertaken along the Mangrol Coast in the Junagadh district for the purpose of sample collection of seaweeds. Notably, in Loej village (Latitude: 21°09'22.9"N; Longitude: 70°02'54.6"E), successful identification and acquisition of brown seaweeds named *Sargassum muticum*, commonly known as Japanese wireweed or japweed were achieved, crucial for the research at hand.[8]

Preservation of Sample

After collection, the next step in the process involves sample processing. The gathered seaweeds undergo multiple rinses with fresh water to eliminate any debris and sand particles. Subsequently, the seaweed is dried in shaded areas. Following drying, the specimens are carefully stored in airtight plastic bags with silica gel for preservation purposes.

MATERIALS & METHODS

Extraction of Sodium Alginate

To facilitate the penetration of reagents during pre-treatment, as well as acid and alkali treatments, the dried *Sargassum muticum* was ground to increase its surface area. The powdered seaweed was then subjected to pre-treatment with 40% formaldehyde for 6 to 12 hours, duration depending on the specific species.

During this process, formaldehyde reacts with phenolic compounds to soften the tissue, promote polymerization, and facilitate the removal of colored pigments by binding with them. As a result of the formaldehyde pre-treatment, the sodium alginate obtained in this study exhibited a light brown coloration.[9]

After washing the samples thrice with distilled water, the pre-treated Sargassum muticum underwent additional treatment with either 0.1N HCl or 1% H_2SO_4 for 24 hours (referred to as "acid wash"). This step aims to increase the solubility of alginate in an alkaline solution, eliminate external salts, residual formaldehyde, and undesired compounds, and create an acidic environment conducive to the conversion of alginate salts.[9,10]

Following the acid treatment, the seaweed was rinsed with distilled water to remove the acid. Subsequently, the seaweed underwent an alkaline treatment with Na_2CO_3 under heat to convert insoluble alginic acid into water-soluble Na-alginate. The resulting mixture was centrifuged to separate and retrieve the sodium alginate, which was then filtered through muslin cloth to obtain the crude alginate extract. Next, the supernatant was precipitated by adding three volumes of 90% ethanol, as sodium alginate does not dissolve well in alcohol-water mixtures.

Finally, the sodium alginate extract underwent purification twice with ethanol, followed by methanol and

acetone, before being dried at room temperature. The yield of alginate was expressed as a percentage of the weight.

$$\text{Yield (\%)} = \frac{Weight\ of\ extracted\ dried\ Na - Alginate}{Weight\ of\ dry\ powdered\ Seaweeds} * 100$$



Fig. 4- Dry Powdered Sodium Alginate

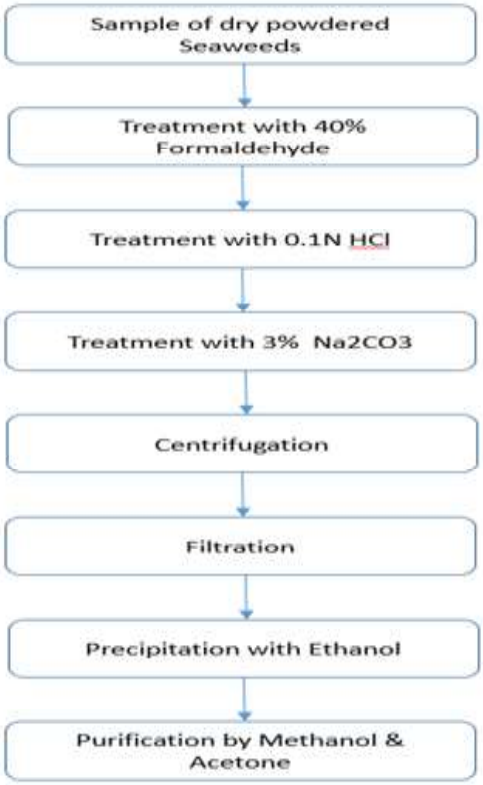


Fig. 5- Flow Diagram for Extraction of Sodium Alginate Process

Preparation of Bioplastic Film at Lab Scale

Bioplastic films are prepared by Film casting method. Three steps can be distinguished in the bioplastic film fabrication process where Alginate samples were mixed with solvents, additives, and plasticizers (water, starch, and glycerol, respectively) in various proportions to produce this transparent and flexible bioplastic.

Compounding & Homogenization

During the compounding and homogenization process, the extracted biopolymer is combined with starch and glycerol. Starch and sodium alginate are gelatinized and homogenized in distilled water using an overhead stirrer in a water bath set at 90°C. Glycerol is then introduced into the film-forming solution and mixed thoroughly. Glycerol serves as a commonly employed plasticizer for hydrophilic biopolymers like alginate, enhancing their mechanical properties, including water permeability and thermal resistance. Cellulose is incorporated as a reinforcing filler to impart mechanical strength properties to the film, such as stiffness and tensile strength.

Casting

In the casting process, the homogeneous, gelatinized mixture is poured onto a flat surface such as a non-stick mould, Teflon sheet, or Petri dish. It is then left to dry in an oven for 24 hours at a controlled temperature of 50°C.

Cross-Linking

The crosslinking step included spraying a solution of calcium chloride ($CaCl_2$) onto the casted solution in order to improve the mechanical properties, moisture sensitivity and visual appearance of the bioplastic, including homogeneity and thickness. Calcium ions in combination with glycerol are claimed to be an optimum combination to enhance the mechanical properties of the alginate-based bioplastic films to increase their water resistance and flexibility. Finally, the film fabrication process is concluded by a drying step, in which the water content of the film is reduced to around 20-30 %.

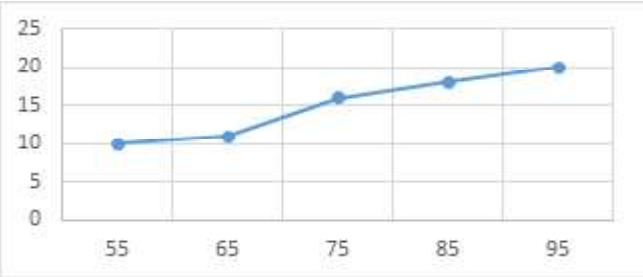


Fig. 6- Bioplastic film

RESULT & DISCUSSION

Effect of Temperature (°C) on Alginate Yield

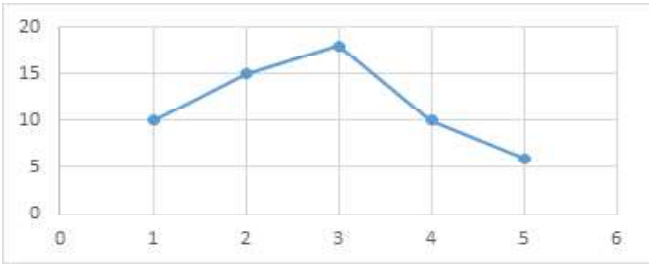
To extract alginate from *Sargassum muticum* using Na_2CO_3, the seaweed was heated to varying temperatures to assess their effect on alginate yield. The experiment covered temperature ranges from 55°C to 95°C, with other variables such as Na_2CO_3 concentration (3%) and extraction time (3 hours) held constant. Results showed that alginate yield was relatively low, ranging from 10% to 12%, at temperatures between 55°C and 65°C. However, the yield increased notably, reaching 15% to 20%, within the temperature range of 75°C to 85°C. The highest yield of 22% was achieved at 95°C. Overall, the maximum yield of alginate extraction from *Sargassum muticum* was observed when temperatures ranged from 90°C to 100°C.[11] While higher extraction temperatures typically lead to increased yields of alginate extraction, excessively high temperatures can cause degradation of alginate conformation, resulting in a decrease in the yield of extracted alginate.[12]



Graph 1- Effect of Temperature on Alginate Yield (%)

Effect of Alkali (Na_2CO_3) concentration on Alginate Yield

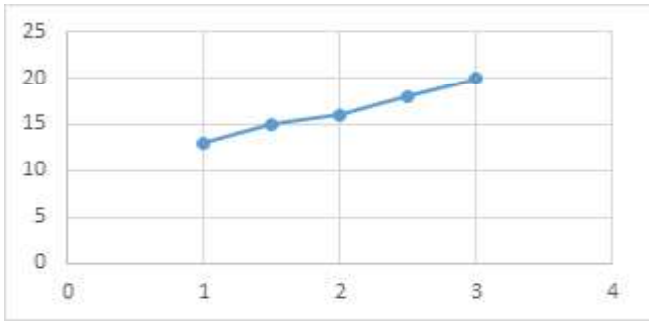
The concentrations of alkali were varied from 1% to 5%, while keeping other process variables constant at 95°C for a period of 3 hours. According to Graph-2, the yield of crude alginate showed a gradual increase from 9% to 18% as the Na_2CO_3 concentration increased from 1% to 3%. However, the yield began to decrease after reaching this level. The highest yield of crude alginate was observed at 3% Na_2CO_3 concentration, while the lowest yield was found at 1%. This trend is attributed to the depolymerization of alginate conformation. The use of ethanol for precipitation in the process also influenced the yield, as ethanol has the ability to bind water from the alginate solution, separating the sodium alginate.[12]



Graph 2- Effect of Alkali concentration on Alginate Yield (%)

Effect of Extraction Time (hours) on Alginate Yield

The yield of alginate is additionally affected by the extraction time. In the experiment, the extraction time ranged from 1 hour to 3 hours, while keeping other parameters constant at 3% Na_2CO_3 concentration and a temperature of 95°C. Alginate yield started at 13% during the initial 1 hour and steadily increased, reaching 20% after 3 hours. Graph-3 illustrates that the maximum amount of alginate yield was attained after 3 hours of extraction time.



Graph 3- Effect of extraction time on Alginate Yield (%)

Advancements and Future Prospects of Sodium Alginate

Pharmaceutical Utility

Sodium alginate plays a vital role in pharmaceutical formulations, serving as a key ingredient. It functions as a thickening agent and imparts sustained release properties to medications.

Due to its high biological tolerability, sodium alginate is well-suited for formulating proteins, peptides, and other biological products.

Sodium alginate's compatibility with aqueous systems and its capacity to cross-link with various agents provide versatile polymeric systems for drug delivery applications.[13,14]

Biomedical Applications

Sodium alginate finds utility in various applications such as wound healing, drug delivery, in vitro cell culture, and tissue engineering.

Its biocompatibility, ability to undergo mild gelation conditions, and ease of modification render sodium alginate ideal for biomedical applications.

Sodium alginate is clinically utilized in various applications, including wound healing dressings, islet transplantation, and chondrocyte transplantation.[14-15]

Future Perspectives

The ability to dynamically control drug delivery from sodium alginate gels in response to external cues holds promise for personalized medicine.

Progress in tissue engineering demands customized sodium alginate gels with precise control over cell-interactive features and adhesion ligands.

Genetic engineering techniques present opportunities to engineer novel sodium alginate polymers with enhanced properties tailored for specific biomedical applications.[13,16]

Physio-chemical Characteristics of Bioplastic Film

Biodegradation

The biodegradation test assesses the degree of damage to bioplastics by measuring the reduction in molecular weight of macromolecules. This test typically involves burying the samples in soil, known as the soil burial test technique. The procedure begins with weighing the initial mass of the sample (m0) before burial. The sample is then placed in a soil-filled pot and left exposed to air for seven days. After the burial period, the bioplastic samples are removed, dried, and re-weighed (m1). The percentage average mass reduction from buried bioplastics is calculated using the following equation:

$$\text{m}\ (\%) = \frac{m1 - m0}{m0} * 100$$

It is evident that synthetic plastics exhibit resistance to degradation, unlike bioplastics containing alginate, which undergo rapid degradation within 15 days. This discrepancy arises from the complex nature of degradation in synthetic plastics. Biodegradability refers to the capacity of polymers to undergo structural changes as a result of the activity of naturally occurring microorganisms such as fungi and bacteria.[11]

Moisture Content

Moisture content analysis was conducted to assess how the affinity of alginate-based films for water was affected by crosslinking with different concentrations of $CaCl_2$. The results showed that cross-linked films exhibited lower moisture content values, indicating that water affinity in alginate films is significantly influenced by crosslinking. This phenomenon can be attributed to the crosslinking process, wherein the COO- groups in alginate bind with Ca_2+ ions. As a result, the alginate chains become less available to bind with H_2O molecules, as the hydrophilic sites along the alginate chains become less exposed, leading to reduced moisture content values.[11]

Water Solubility

The solubility of the film is assessed through a water solubility test, with the solubility expressed as a percentage. The difference between the initial weight and the final weight after soaking and drying is recorded. The solubility (S%) is calculated using the following equation:

(S%) = [(W1-W2)/W1] *100

W1 - initial weight of the film

W2 - final weight of the film after soaking in water and drying in the oven

S% - solubility of the film

CONCLUSION

In conclusion, this study underscores the significant potential of seaweeds as a sustainable source for bioplastics production. As concerns about the environmental impact of traditional plastics continue to rise, the necessity for eco-friendly alternatives has become paramount. Seaweeds offer a promising solution due to their abundant biomass, ease of cultivation, and biodegradability. Additionally, seaweed-based bioplastics possess desirable characteristics such as resistance to microwave radiation, durability, and flexibility, rendering them suitable for diverse applications including food packaging. By synthesizing biodegradable bioplastics from seaweeds, we can play a role in mitigating plastic pollution while reducing the strain on natural resources. Overall, this review highlights the importance and advantages of embracing seaweeds as a sustainable alternative for plastic production, paving the way for a greener and more environmentally conscious future.

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